Roadmap. §10 runs the φ^3 four-point function through LSZ: disconnected pieces give no scattering, and the connected tree diagrams give $\langle f|i\rangle = (2\pi)^4\delta^4(k_{\text{in}} - k_{\text{out}})\, iT$. From this example come the momentum-space Feynman rules for iT at any order. Bold $\mathbf{\Delta}$: exact propagator; plain Δ: free. Next: cross sections from T (LecNote10, §11); $2 \to 2$ at one loop (LecNote14, §20). Supplements: AsymptoticStates, SymmetryFactors, FeynmanRulesLSZ.

Sec 10. Scattering Amplitudes & Feynman Rules.

With $Z(J) = e^{iW(J)}$

we can calculate time-ordered amplitudes through functional derivatives.

$\frac{1}{i}\frac{\delta}{\delta J}\; i\Delta\; \frac{1}{i}\frac{\delta}{\delta J}$

$\frac{1}{i}\mathbf{\Delta}(x_1 - x_2) \equiv \langle 0|\, T\,\varphi(x_1)\,\varphi(x_2)\,|0\rangle$. with $\delta_j = \frac{1}{i}\frac{\delta}{\delta J(x_j)}$

$= \delta_1\delta_2\, Z(J)\big|_{J=0} = \delta_1\delta_2\, iW(J)\big|_{J=0} - \delta_1 iW(J)\big|_{J=0}\,\delta_2 iW(J)\big|_{J=0}$

$= \delta_1\delta_2\, iW(J)\big|_{J=0}$

as $\delta_j W(J)\big|_{J=0} = \langle 0|\varphi(x_j)|0\rangle = 0$

δ_1 removes a source and gives a label x_1

$\frac{1}{i}\Delta(x_1 - x_2)$ is given by all diagrams with two sources

at lowest order 

Thus we get

Feynman Propagator $\Delta(x-x') = \int \frac{d^4k}{(2\pi)^4} \frac{e^{-ik(x-x')}}{k^2+m^2-i\varepsilon}$

$\frac{1}{i}\mathbf{\Delta}(x_1 - x_2) = \frac{1}{i}\Delta(x_1 - x_2) + O(g^2)$

$\langle 0|\, T\,\varphi(x_1)\varphi(x_2)\varphi(x_3)\varphi(x_4)|0\rangle = \delta_1\delta_2\delta_3\delta_4\, Z(J)\big|_{J=0}$

$= [\delta_1\delta_2\delta_3\delta_4\, iW(J) + (\delta_4\delta_2 iW(J))(\delta_3\delta_4 iW(J))$
$+ (\delta_1\delta_3 iW(J))(\delta_2\delta_4 iW(J))$
$+ (\delta_1\delta_4 iW(J))(\delta_2\delta_3 iW(J))]_{J=0}$

$= \delta_1\delta_2\delta_3\delta_4\, iW(J)\big|_{J=0} + \frac{1}{i}\Delta(x_1 - x_2)\frac{1}{i}\Delta(x_3 - x_4) + \dots$

1 Check: $x_1 = X + \frac{1}{2}x_{11'}$, $x_1' = X - \frac{1}{2}x_{11'}$ make the exponent $i(k_1 - k_1')X + i\bar{k}_{11'}x_{11'}$, so $\int d^4X$ gives $(2\pi)^4\delta^4(k_1 - k_1')$. The $\Delta(x_1 - x_2)\Delta(x_1' - x_2')$ term gives $\delta^4(k_1 + k_2) = 0$ ($k_1^0 + k_2^0 > 0$). So each disconnected term needs some $k' = k$: no scattering, excluded by taking every $k' \neq$ every k (LecNote4 p. 3).

We can insert these terms to the LSZ reduction for 2→2 process

$$\langle f|i\rangle = i^4 \int d^4x_1\, d^4x_2\, d^4x_3\, d^4x_4\, e^{-i(k_1x_1 + k_2x_2 - k_1'x_1' - k_2'x_2')}$$

$$(-\partial_1^2 + m^2)(-\partial_2^2 + m^2)(-\partial_3^2 + m^2)(-\partial_4 + m^2)$$

$$\langle 0|T\varphi(x_1)\varphi(x_2)\varphi(x_1')\varphi(x_2')|0\rangle.$$

There are several products of Green's functions contributing,

let's check $\frac{1}{i}\Delta(x_1 - x_1')\,\frac{1}{i}\Delta(x_2 - x_2')$

$$\int d^4x_1 d^4x_2 d^4x_1' d^4x_2'\, e^{-i(k_1x_1 + k_2x_2 - k_1'x_1' - k_2'x_2')} F(x_{11'})F(x_{22'})$$

where we defined $F(x_{ij}) = (-\partial_i^2 + m^2)(-\partial_j^2 + m^2)\Delta(x_{ij})$.

$$x_{ij} = x_i - x_j,\quad \bar{k}_{ij} = (k_i + k_j)/2.$$

$$= (2\pi)^4\delta^4(k_1 - k_1')(2\pi)^4\delta^4(k_2 - k_2')\,\tilde{F}(\bar{k}_{11'})\,\tilde{F}(\bar{k}_{22'})$$

$\tilde{F}(k)$ is ~~is~~ $F(x_{ij})$'s **Fourier Transform**. check ①

but clearly this represents no scattering as $k_1 = k_1'$, $k_2 = k_2'$

In general, we are only interested in diagrams where something happened (scattering), which is "sufficiently" represented by

Fully Connected diagrams.

what if some connected, some not? They can be factorized.

1 3 groups = 3 channels: which of x_2, x_1', x_2' shares x_1's vertex. The parenthesis lists the swaps behind $8 = 2 \cdot 2 \cdot 2$ (A↔B, C↔D, the two vertices): the symmetries of the sourced diagram, so $S = 8$. With labelled ends a tree keeps none (each vertex lies on the path between two labelled ends); loops can (rule 8, p. 7).

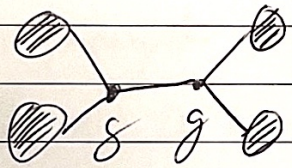
From here on, we restrict ourselves to those diagrams alone and define Connected correlation functions via.

$$\langle 0|T\varphi(x_1)\cdots\varphi(x_E)|0\rangle_C \equiv \delta_1\cdots\delta_E\, iW[J]|_{J=0}.$$

Back to 2→2 process.

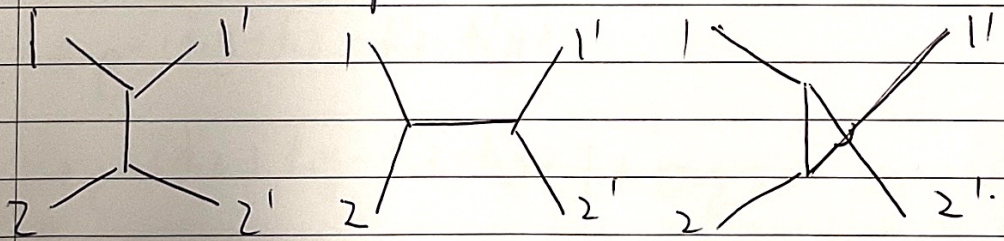
$$\langle 0|T\varphi(x_1)\varphi(x_2)\varphi(x_1')\varphi(x_2')|0\rangle_C = \delta_1\delta_2\delta_1'\delta_2' iW[J]|_{J=0}$$

at the lowest order in g, the contributing process is



with the 4 ⊗ (sources) functional derived away by δ.

There are 4! ways to $\delta_1\delta_2\delta_3\delta_4$.
which are 3 groups (A&B pairing, C&D pairing, and/or each with 8 identical exchange of the 4 vertices reflection diagrams. ①



Note here that the factor 8 nicely cancels the symmetry factor $S=8$ of the original sourced diagram.

This is a general result for TREE-level diagrams (those without closed loops), that the symmetry factor got cancelled with explicit labelling. This is the first non-zero contribution to the amplitude.

each propagator is $\frac{1}{i}$ as $\frac{1}{i}\frac{\delta}{\delta J}\, i\Delta \cdot \frac{1}{i}\frac{\delta}{\delta J}$.

Now.

$$\langle 0|T\varphi(x_1)\varphi(x_2)\varphi(x_1')\varphi(x_2')|0\rangle_c$$

$$= (ig)^2 \left(\frac{1}{i}\right)^5 \int d^4y\, d^4z\, \Delta(y-z) \cdot X$$

$$[\Delta(x_1-y)\Delta(x_2-y)\Delta(x_1'-z)\Delta(x_2'-z)$$

$$+\Delta(x_1-y)\Delta(x_2-z)\Delta(x_1'-z)\Delta(x_2'-y)$$

$$+\Delta(x_1-z)\Delta(x_2-y)\Delta(x_1'-z)\Delta(x_2'-y)] + O(g^4).$$

Put in LSZ. Each Klein–Gordon wave operator acts on a propagator to give.

$$(-\partial_i^2 + m^2)\,\Delta(x_i-y) = \delta^4(x_i-y) \leftarrow$$

(note that Δ is free-particle propagator, as it is from part $Z_0(J)$).

Then we get.

$$\langle f|i\rangle = (ig)^2 \frac{1}{i} \int d^4y\, d^4z\, \Delta(y-z)$$

$$[e^{i(k_1y+k_2y-k_1'z-k_2'z)}$$

$$+e^{i(k_1y+k_2z-k_1'z-k_2'y)}$$

$$+e^{i(k_1z+k_2y-k_1'z-k_2'y)}] + O(g^4).$$

knowing that $\Delta(y-z) = \int \frac{d^4k}{(2\pi)^4} \frac{e^{-ik(y-z)}}{k^2+m^2-i\epsilon}$.

we can get

~~$\langle f|i\rangle = -ig^2 \int d^4y$~~ e

$$\langle f|i\rangle = -ig^2 \int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2+m^2-i\epsilon}$$

$$\times [(2\pi)^4 \delta^4(k_1+k_2+k) + (2\pi)^4 \delta^4(k_1'+k_2'+k)$$

$$+ (2\pi)^4 \delta^4(k_1-k_1'+k)(2\pi)^4 \delta^4(k_2-k_2'+k)$$

$$+ (2\pi)^4 \delta^4(k_1-k_2'+k)(2\pi)^4 \delta^4(k_2-k_1'+k)] + O(g^4).$$

$$= ig^2 \cdot (2\pi)^4 \cdot \delta^4(k_1+k_2-k_1'-k_2')$$

$$\left[\frac{1}{(k_1+k_2)^2+m^2-i\epsilon} + \frac{1}{(k_1-k_1')^2+m^2-i\epsilon} + \frac{1}{(k_1-k_2')^2+m^2-i\epsilon}\right] + O(g^4).$$

The overall Dirac delta shows the energy momentum conservation. So

$$\langle f|i\rangle = (2\pi)^4 \delta^4(k_{in}^{total} - k_{out}^{total})\, iT$$

↑ the scattering Matrix Element.

Bring QFT to dummies.

Now we can finally arrive at the Feynman Rules that we learned from the above process for calculating iT.

The Feynman Rules are: (2)

1 Should read $\delta^4(k_1 - k_1' + k)\,\delta^4(k_2 - k_2' - k)$ and $\delta^4(k_1 - k_2' + k)\,\delta^4(k_2 - k_1' - k)$: $\Delta(y-z)$ carries k into one vertex and out of the other. As written, $\int d^4k$ would not give $\delta^4(k_1 + k_2 - k_1' - k_2')$. The final line is correct.

2 Why these rules: put $\Delta(x-y) = \int \frac{d^4k}{(2\pi)^4} e^{-ik(x-y)} \tilde{\Delta}(k)$ in every line. Each vertex $ig\int d^4y$ gives $ig(2\pi)^4\delta^4(\sum k)$: rule 5. Each internal line keeps $\int \frac{d^4k}{(2\pi)^4}$ and $\frac{1}{i}\tilde{\Delta} = \frac{-i}{k^2+m^2-i\epsilon}$. LSZ's $i(-\partial^2+m^2)$ turns each external $\frac{1}{i}\Delta$ into δ^4: factor 1. Unfixed k's are loops (rule 7); one δ^4 remains: strip it off to get iT.

① Draw lines (external lines) for each incoming or outgoing particles

② Leave one end of each external line free and connect the other end to a vertex where exactly three lines meet.
for ϕ^3 theory.

Include extra internal lines to do this if necessary.

Draw all diagrams that are Topologically Inequivalent.

③ On each incoming line, draw arrow towards the vertex
On each outgoing line, draw arrow away from the vertex
On each internal line, draw it arbitrarily.

④. Assign each line its own momentum. The four momentum of external lines are their own momentum.

⑤. Thinking of the four-momenta as following along the arrows, and conserve four-momenta at each vertex.
(For a tree diagram, this fix all momenta)

⑥. The value of each diagram contains the following factors
each external line has, 1;
each internal line with momentum k, $\frac{-i}{k^2+m^2-i\epsilon}$.
each vertex, $iZ_g g$.

7 (loop). A diagram with L ~~loop~~ closed loops will have L internal momenta unfixed by rule #5. Integrate over these 4-momenta with measure

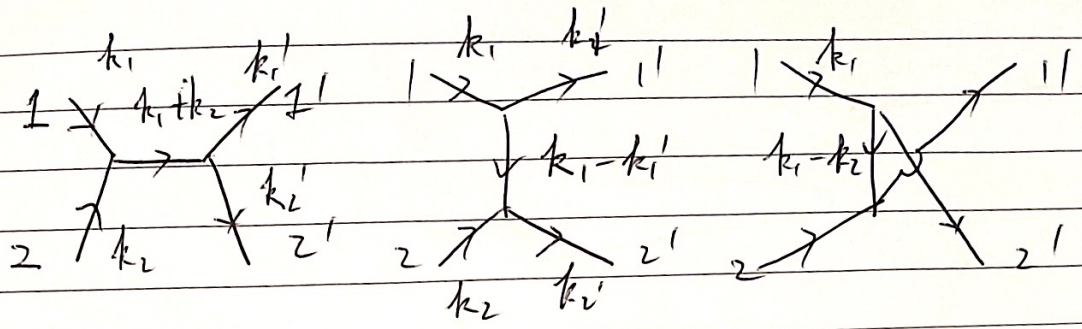
$$\frac{d^4 l_i}{(2\pi)^4}.$$

8. ~~A~~ (loop) A loop diagram may have left over ~~symmetry factors~~ if ~~there are~~ exchange of internal propagators and vertices that leave the diagram unchanged. Divide the value by such ~~symm~~ leftover symmetry factors.

9. (counter terms) Include diagrams with the counterterm vertex that connects two propagators, each with the same four momentum k. The value of ~~this vertex~~ is $-i(Ak^2 + Bm^2)$, where $A \equiv Z_\varphi - 1$ and $B = Z_m - 1$, each are expected to be $O(g^2)$.

10. The value of $i\Gamma$ is given by ~~a sum over~~ the values of all these diagrams [1]

1 Not all diagrams: never draw tadpoles (Y cancels them) or loops on external legs. A leg with all its loops sums to the exact $\tilde{\Delta}$, whose pole sits at the physical m with residue 1, as LSZ assumed; Z_φ, Z_m enforce this (LecNote12). So $i(k^2+m^2)\frac{1}{i}\tilde{\Delta} \to 1$: rule 6's factor 1. $Z_g = 1 + O(g^2)$ (LecNote13): tree vertex ig.



At tree-level, we have.

$$iT = (-ig)^2 \left(\frac{-i}{(k_1+k_2)^2+m^2-i\epsilon} + \frac{-i}{(k_1-k_1')^2+m^2-i\epsilon} + \frac{-i}{(k_1-k_2')^2+m^2-i\epsilon} \right) + O(g^4)$$

Now we can proceed to get observable cross section from the scattering amplitudes from iT.

1 With $s = -(k_1+k_2)^2$, $t = -(k_1-k_1')^2$, $u = -(k_1-k_2')^2$: $T = g^2\left[\frac{1}{m^2-s} + \frac{1}{m^2-t} + \frac{1}{m^2-u}\right]$. Physically $s \geq 4m^2$, $t, u \leq 0$: no denominator vanishes, $i\epsilon$ drops, T is real (every tree is $i^V(-i)^{V-1} = i$ times real). T is symmetric in s, t, u (crossing); $t \leftrightarrow u$ is $1' \leftrightarrow 2'$. $O(g^4)$: one loop (LecNote14).